

★ WEEK 8 · PART 3 of 3

Midterm Synthesis

Seven Weeks, One Connected Story

The Part I thread · a model decision tree · sample questions · exam tips

Part I was not seven disconnected topics — it was one argument. This synthesis retraces that thread, organizes the toolbox, and prepares you for the midterm on Weeks 1-7.

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PART 3 · MIDTERM SYNTHESIS

Tying Part I Together

The connected story through Weeks 1-7, a decision tree for choosing a model, a review checklist, worked sample questions, and how to prepare for the exam.

What This Synthesis Is For

Review with understanding, not memorization

1

Part I was one argument

From a single return to a traded spread, each week answered a question the previous one raised. The midterm tests whether you see that thread.

2

Know the concept AND the trigger

Exams ask 'which tool for THIS data?'. So for every model, know what it does and the data pattern that calls for it.

3

This deck retraces and organizes

A roadmap, a decision tree, a checklist, and worked questions — everything you need to consolidate Weeks 1-7.

Part 3 in one line: understand the connected story, and you can answer anything on Weeks 1-7.

One Connected Story

How the seven weeks fit together

Model the MEAN and find it nearly unpredictable; so model the VARIANCE; then model series JOINTLY; and finally their LONG-RUN equilibrium. Each step follows from the limits of the last.

The logical chain

- W1-4: model the mean — returns barely have one.
- W5: so model volatility, where the structure lives.
- W6-7: move to many series, then their equilibrium.

Why it matters for the exam

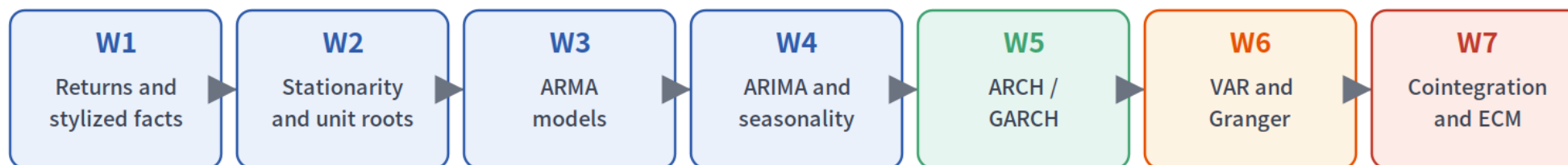
- Questions test transitions, not just definitions.
- 'Why move from ARIMA to GARCH?' is a classic.
- The thread IS the answer key.

Part I: One Connected Story

Model the mean, discover it's unpredictable, model the variance, go multivariate, then capture the long-run equilibrium — with the pairs trade threading through it all.

Part I: One Connected Story

Each week answered a question the previous one raised - from a single return to a traded spread.



The logic that connects them

Model the MEAN (W1-4) - find it is nearly unpredictable - so model the VARIANCE (W5) - then model series JOINTLY (W6) - and finally their LONG-RUN equilibrium (W7). Part II now unifies all of this in state space.

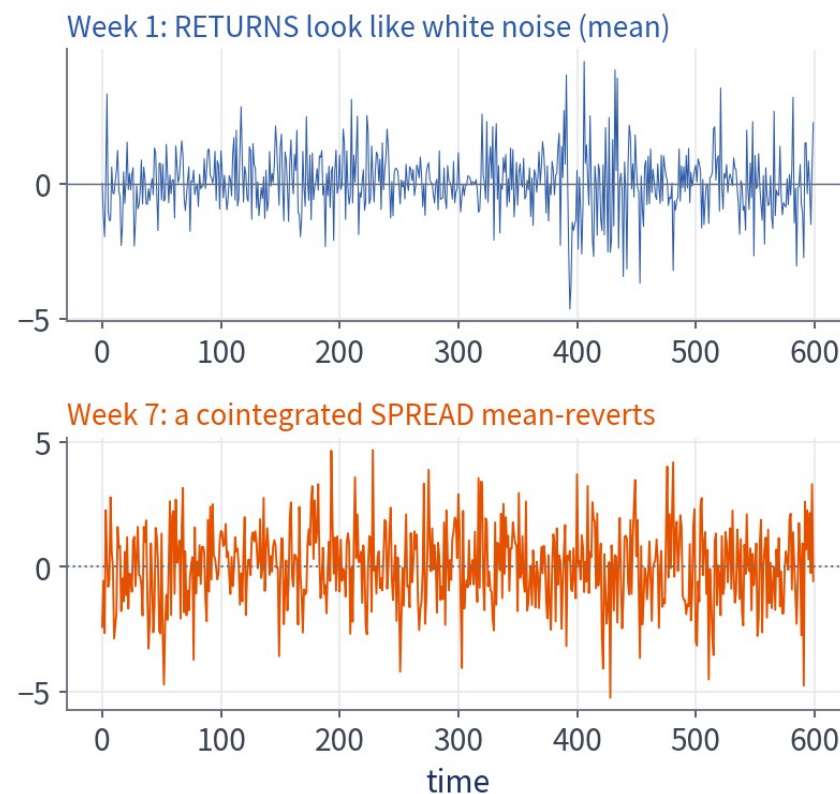
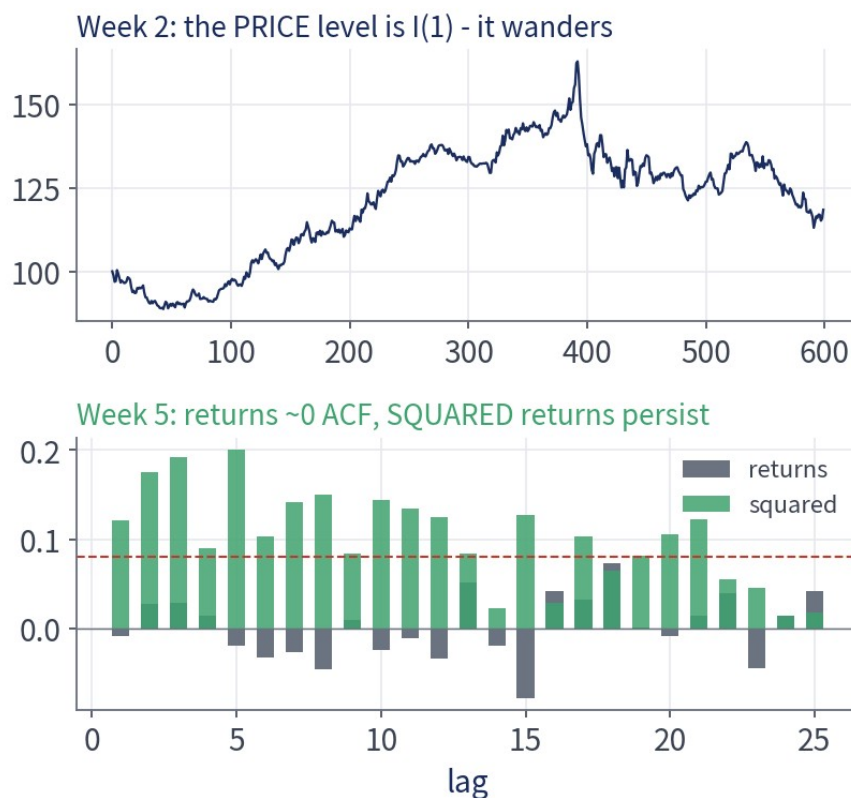
The pairs-trading thread ties it together

Stationarity (W2) - volatility bands (W5) - cointegration anchor (W7) - and, in Part II, an adaptive hedge ratio. If you can trace the pair through all seven weeks, you understand Part I.

One Dataset, Seven Lenses

The same data reads differently through each week's tool: a wandering price (W2), noise-like returns (W1), clustering in squared returns (W5), and a reverting spread (W7).

One dataset, seven lenses: each week of Part I reads the same data differently



Weeks 1-2: Returns and Stationarity

The foundation everything rests on

Key ideas

- Returns, not prices; the stylized facts (fat tails, clustering).
- Stationarity: constant mean and variance over time.
- Unit roots and the $I(0)$ vs $I(1)$ distinction (ADF/KPSS).

When it's tested

- Any question starts here: is the series stationary?
- Spurious regression: high R-squared between $I(1)$ series.
- The answer routes every later modeling choice.

Weeks 3-4: ARMA and ARIMA

Modeling the mean

Key ideas

- ARMA for stationary series; ACF/PACF to identify p and q .
- ARIMA differences d times first; random walk = ARIMA(0,1,0).
- SARIMA adds seasonal terms at lag s (the airline model).

When it's tested

- Identify orders from an ACF/PACF description.
- Choose d from unit-root behavior.
- The humbling fact: returns have almost no ARMA structure.

Week 5: ARCH and GARCH

Modeling the variance

Key ideas

- Conditional vs unconditional variance; volatility clustering.
- GARCH(1,1) with persistence $\alpha + \beta$; mean reversion.
- Student-t for fat tails; GJR/EGARCH for leverage; VaR.

When it's tested

- Flat return ACF but persistent SQUARED-return ACF $\rightarrow$ GARCH.
- Asymmetry after drops $\rightarrow$ GJR/EGARCH.
- Turning a volatility forecast into VaR.

Week 6: VAR and Granger

Modeling series jointly

Key ideas

- VAR: each series on the past of all series; stability, lag order.
- Granger causality tests PREDICTION, not true causation.
- IRF traces shocks; FEVD budgets variance (needs orthogonalization).

When it's tested

- 'Does X Granger-cause Y' — and the causality caveat.
- Reading an IRF or FEVD plot.
- Why ordering matters for Cholesky-based analysis.

Week 7: Cointegration and ECM

The long-run equilibrium

Key ideas

- Two $I(1)$ series with a stationary combination — cointegrated.
- Test with Engle-Granger or Johansen (the rank).
- Error correction: the VECM, speed of adjustment, half-life.

When it's tested

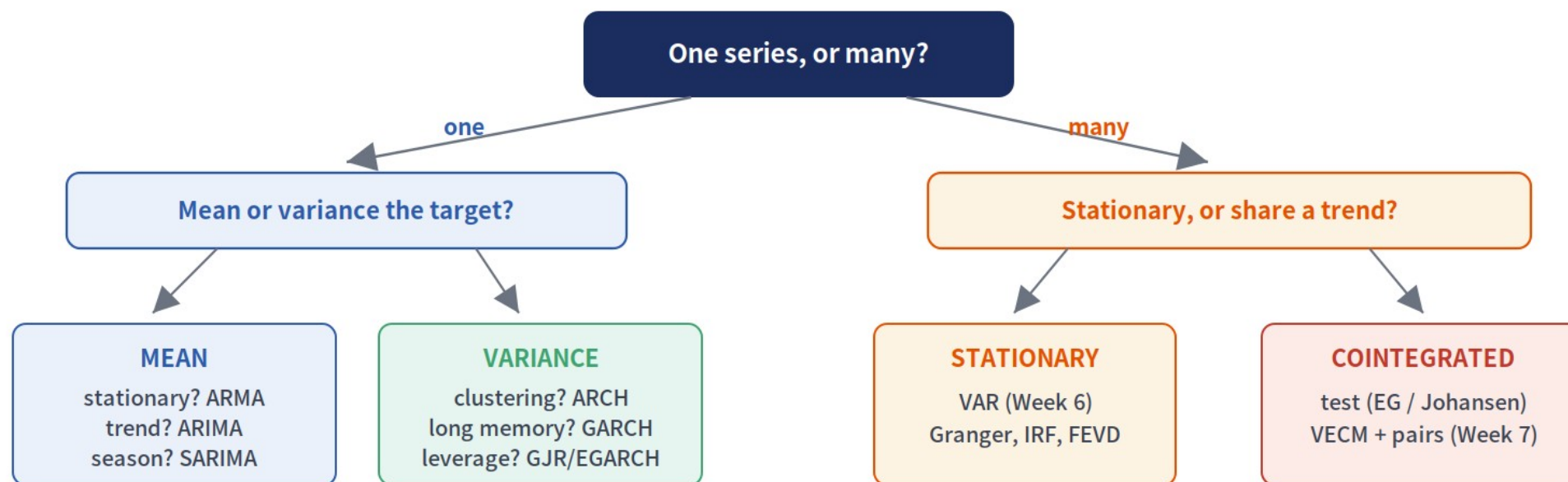
- Spurious vs cointegrated: is the residual stationary?
- Reading alpha (adjustment speed) and the half-life.
- Pairs trading built on the spread — and its breakdown risk.

Which Model, When?

One series or many? Mean or variance? Stationary or sharing a trend? The tree routes any dataset to the right tool — and it always begins with a stationarity check.

Which Model, When?

A decision tree over the Part I toolbox.



Always start by checking stationarity (Week 2) - it routes the entire decision.

The Toolbox on Two Axes

What you model, and how many series

Target	Single series	Multiple series
Mean / level	ARMA, ARIMA, SARIMA	VAR
Variance / risk	ARCH, GARCH, GJR/EGARCH	(multivariate GARCH — later)
Long-run link	unit roots, differencing	cointegration, VECM

Almost every Part I model lives in one of these cells — locate the data, and you locate the tool.

Key Equations: Mean Models

The ones worth memorizing

ARMA(p, q) : $\phi(L)X_t = \theta(L)\varepsilon_t$ stationary if roots of ϕ outside unit circle

ARIMA(p, d, q) : $\phi(L)(1 - L)^d X_t = \theta(L)\varepsilon_t$ RW = ARIMA(0, 1, 0)

SARIMA(p, d, q)(P, D, Q) $_s$: add seasonal terms at lag s

What to know

- The lag-operator form, and what p, d, q each control.
- Stationarity: AR roots outside the unit circle.
- Random walk = ARIMA(0,1,0); seasonal terms at lag s .

How it is tested

- Map an ACF/PACF pattern to orders.
- State the differencing needed for a trend.
- Recognize special cases by name.

Key Equations: Risk and Multivariate

The rest of the must-knows

$$\begin{aligned}\text{GARCH}(1,1) : \sigma_t^2 &= \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 && \text{persistence} = \alpha + \beta \\ \text{VAR}(p) : \mathbf{Y}_t &= \mathbf{c} + A_1 \mathbf{Y}_{t-1} + \cdots + A_p \mathbf{Y}_{t-p} + \boldsymbol{\varepsilon}_t \\ \text{Cointegration} : X_t, Y_t &\sim I(1), \quad z_t = Y_t - \beta X_t \sim I(0)\end{aligned}$$

What to know

- GARCH(1,1) and persistence alpha + beta.
- The VAR form and its cross-terms.
- The cointegration condition: a stationary spread of I(1) series.

How it is tested

- Interpret alpha + beta near 1 (persistent volatility).
- Read off-diagonal VAR terms as lead-lag.
- Identify cointegration from a stationary residual.

The Pairs-Trading Thread

One example that touches every week

If you can trace the pairs trade through Part I, you understand the whole. Stationarity finds the pair; volatility sizes the bands; cointegration anchors it; error correction is why it reverts.

Week by week

- W2: are the two series $I(1)$? (the prerequisite)
- W5: GARCH bands that scale with volatility.
- W7: cointegration test, spread, and error-correction speed.

The exam angle

- A single pairs scenario can test five weeks at once.
- Expect 'what would you check next?' style questions.
- The thread gives you the order of operations.

Weeks 1-7 at a Glance

Concept and trigger, side by side

Week	Core concept	Data pattern that calls for it
1-2	Returns, stationarity, unit roots	always first — decide $I(0)$ vs $I(1)$
3-4	ARMA / ARIMA / SARIMA	a stationary (or differenced) mean, maybe seasonal
5	ARCH / GARCH	flat return ACF, clustered squared returns
6	VAR, Granger, IRF/FEVD	several interacting stationary series
7	Cointegration, ECM	$I(1)$ series sharing a long-run equilibrium

If you can fill the third column from a data description, you are ready.

Common Exam Traps

The mistakes that cost easy marks

The classic errors

- Regressing $I(1)$ levels and trusting a high R-squared.
- Reading Granger causality as true causation.
- Fitting ARIMA to returns expecting a tradable signal.

The right reflexes

- Check stationarity first — always.
- Say 'predicts', not 'causes', for Granger.
- Returns are for VARIANCE modeling, not mean forecasting.

A Regression That Looks Too Good

Weeks 2 and 7

You regress one stock's price on another's and get an R-squared of 0.95. A colleague concludes they are strongly related and wants to trade the relationship. State your concern and the test that settles it.

Think about

- Are the two price series stationary, or $I(1)$?
- What does a high R-squared between $I(1)$ series usually mean?
- How do you tell a real relationship from a spurious one?

The trap

- A high R-squared alone proves nothing here.
- It is the classic spurious-regression warning.
- The answer is on the next slide.

Spurious, Unless Cointegrated

The worked solution

The concern

- Both prices are almost certainly $I(1)$ (unit roots).
- Regressing $I(1)$ on $I(1)$ gives a spuriously high R-squared (Week 2).
- So the 0.95 is likely a statistical illusion, not a relationship.

The test that settles it

- Confirm each series is $I(1)$ (ADF/KPSS).
- Test the regression RESIDUAL for stationarity (Engle-Granger).
- Stationary residual $\rightarrow$ cointegrated (real, tradable); if not $\rightarrow$ spurious.

Reading a Return Series

Weeks 1, 3, and 5

A daily equity return series has an ACF near zero at all lags, but its SQUARED returns are strongly autocorrelated, and volatility jumps more after price drops than after rises. Which model would you fit, and why?

Think about

- What does a flat return ACF tell you about the mean?
- What does autocorrelation in squared returns indicate?
- What does the asymmetry after drops imply?

The clues

- Three separate hints point to three modeling choices.
- Each maps to a specific tool from Week 5.
- The answer is on the next slide.

A GARCH With Leverage and Fat Tails

The worked solution

Reading the clues

- Flat return ACF → the mean is unpredictable; don't fit ARIMA.
- Autocorrelated squared returns → volatility clustering (ARCH effects).
- Bigger jumps after drops → the leverage effect.

The model

- A GARCH-family model for the variance, with a trivial mean.
- GJR-GARCH or EGARCH to capture the asymmetry.
- Student-t innovations for the fat tails.

Format, Coverage, and Tips

What to expect and how to prepare

Coverage

- All of Weeks 1-7, weighted toward the connected thread.
- Concept identification, model choice, and interpretation.
- Reading plots (ACF/PACF, IRF, spreads) is fair game.

Tips

- Always state the stationarity check first.
- Justify EVERY model choice with the data pattern.
- Distinguish prediction from causation, spurious from cointegrated.

How to Prepare

Five focused steps

1 · Thread

the story

Retrace why
each week
follows the last.

2 · Tree

model choice

Practice routing
data to a
model.

3 · Plots

read them

Interpret
ACF/PACF, IRF,
and spreads by
eye.

4 · Traps

avoid

Drill spurious vs
cointegrated,
Granger caveat.

5 · Pairs

one example

Trace the pairs
trade through
all seven weeks.

Master the transitions and the decision tree, and the details fall into place.

Are You Ready?

Three questions that test the whole thread

Q1	Given a new series, what is the very first thing you check?	Stationarity ($I(0)$ vs $I(1)$)
Q2	Returns are unpredictable in the mean. So what do you model?	The variance (GARCH)
Q3	Two $I(1)$ series, stationary spread. What framework, and what trade?	Cointegration → pairs

If these come easily, you have the thread — and the midterm.

Three Things to Remember

Week 8 in one screen

1

State space is the unifying framework

Parts 1-2: a hidden state seen through noise, estimated by a predict-update cycle — filtering, smoothing, forecasting, and time-varying parameters, all in one language.

2

Part I is one connected story

Part 3: model the mean, then the variance, then jointly, then the long-run link. The decision tree and the pairs thread tie all seven weeks together.

3

Know the concept AND the trigger

For the midterm, for every tool know what it does and the data pattern that calls for it. Start every answer with the stationarity check.

Next — Week 9: the Kalman filter, the optimal engine behind everything in Part II.